

Atul

New Delhi, India | +918076009041 | atulchoudhary7781@gmail.com | in/atul-choudhary | [GitHub](#) | [Portfolio](#)

SUMMARY

AI Product Engineer specializing in Python-based AI applications, machine learning models, RAG pipelines, and multi-agent orchestration. Experienced in developing and deploying scalable generative AI solutions with PyTorch, Vertex AI, REST APIs, and database integrations, including multimodal autonomous agents and semantic search systems. Brings a focus on model optimization, production reliability, and practical AI infrastructure.

WORK EXPERIENCE

Venura Tech | *Internship*

May 2026 - Jun 2026

- Operate as an advanced specialist in AI training and system operations, refining and optimizing large language models with PyTorch, applying regression techniques, and supporting model evaluation to improve performance.
- Developing and evaluating complex agentic workflows, multi-agent orchestration, and advanced RAG pipelines.
- Engineering high-quality prompt datasets (Few-shot, Chain-of-Thought) to improve model accuracy, reasoning, and reduce latency.
- Collaborate asynchronously with global teams to deploy scalable, production-grade generative AI solutions for enterprise clients through software development and REST APIs built with FastAPI.
- Implemented semantic search capabilities using embeddings to enhance retrieval performance in RAG pipelines.
- Utilize pandas and numpy for data processing, preprocessing, and feature engineering to support machine learning model training and evaluation.

Jarvis AI | *AI Product Engineer*

Feb 2026 - Present

- Design and build Jarvis, a multimodal autonomous AI agent optimized for near-zero latency using Groq SDK, Azure OpenAI, OpenAI API, Claude API, and high-performance RAG pipelines.
- Integrate multi-agent orchestration to handle complex, multi-step reasoning tasks with real-world tool integration and scalable AI workflows.
- Migrate the UI to Flet for a seamless cross-platform experience and architect agentic workflows for autonomous decision-making through REST APIs using Flask.
- Stack: Python; Groq SDK; Flet; LangChain; RAG; Multi-agent Orchestration; Vertex AI
- Integrated chat interface with PostgreSQL database to manage structured data storage and retrieval.

EDUCATION

Indian Institute of Computer Science | *Diploma, Digital Marketing*

Mar 2025 - Jul 2025

Indian Institute of Computer Science | *Diploma, Software Engineering*

2022 - 2025

School of Open Learning (DU SOL), University of Delhi | *Bachelor of Arts (BA)*

2022 - 2025

PROJECTS

Jarvis – Multimodal Autonomous AI Agent | Jarvis – Multimodal Autonomous AI Agent

2026 - Present

- Built a multimodal AI agent capable of voice, text, and vision inputs using Groq SDK for ultra-low latency inference.
- Implemented retrieval-augmented generation pipelines using OpenAI embeddings and semantic search with a vector database to enable long-term memory and context-aware responses.
- Deployed a multi-agent orchestration layer using LangChain agents and prompt engineering to delegate tasks across specialized sub-agents, supporting scalable AI model deployment.
- Technologies: Python, Groq SDK, LangChain, Vertex AI, Google Cloud Platform, Flet, Git, REST APIs.
- Built FastAPI-based RESTful endpoints to integrate the AI agent with web, voice, and vision front ends, enabling scalable deployment and maintainable software development practices.

SKILLS

Languages: Python, JavaScript, Bash, SQL, C#, C++, Go, Java

AI/ML Frameworks: LangChain, Groq SDK, Vertex AI, RAG Pipelines, Prompt Engineering, Agentic Workflows, PyTorch, TensorFlow, Scikit-Learn, Pandas, Numpy

Cloud / Tools: Google Cloud Platform (GCP), Vertex AI Studio, Google Analytics, Flet, Git, Microsoft Azure, Azure OpenAI, REST APIs, PostgreSQL, LLM APIs, OpenAI API, Claude API, FastAPI, Flask, API Development

Specializations: Autonomous Agents, Multi-Agent Orchestration, LLM Optimization, RAG, Few-Shot & Chain-Of-Thought Prompting, Machine Learning, Large Language Models, Natural Language Processing, Embeddings, Semantic Search, Regression, Classification, Clustering

Engineering & Infrastructure: Software Development, Machine Learning Infrastructure, Model Deployment, Model Evaluation, Data Processing, Data Structures And Algorithms, Large-Scale Distributed Systems

CERTIFICATIONS & ACHIEVEMENTS

- Introduction to Generative AI: Google / Vertex AI Studio
- AI for Coding using Tabnine
- Gen AI: Beyond the Chatbot
- Technology Job Simulation: Deloitte Australia